

users carry on their key chains. These tokens generate one-time passwords (OTPs) for the second stage of the login process. However, hardware tokens are comparatively expensive, difficult to track and replace when broken, and the effort for distributing these is time consuming. Also, these are easy to lose and hard to use.

Banks have come up with several solutions for OTP generation such as short message service based, mobile application based, email based, software token based, interactive voice response (IVR) based solutions and so on.

Carry out up to date patching of machines, use antivirus

Do you know Zeus? I am not talking about the Greek god of the sky and thunder. Zeus is a banking Trojan that is being used to steal banking information by keystroke logging and form grabbing. Zeus's mobile variant called ZitMo is well-known to circumvent popular TFA schemes with security codes being provided via text messages.

SpyEye and Carber have developed their respective mobile counterparts. Dyre, which typically targets customers of large financial institutions, was recently used in a large-scale, credential-phishing campaign targeting international banks.

Each malware tries to evade detection by an antivirus. It intercepts keystrokes, browser data, stored files and basically everything to sneak into a banking account and initiate an illegal money transfer. It even tries to install mobile malware on a smartphone, which allows criminals to steal the OTP.

By regularly applying software patches and using an updated genuine antivirus solution, you can stay away from this malicious software to a good extent. In order to have a healthy PC, always ensure that cracked/pirated operating system/software are not installed. Always remember that nothing comes for free.

Do not trust open/free Wi-Fi

Do you pay your bills online while having pizza and enjoying free Wi-Fi at your favourite coffee shop? Better stop before you pay. Like a lion waiting for its prey, someone is waiting in that Wi-Fi network to steal your credit card information, Internet banking credentials and a lot more even before it reaches your bank.

Experts warn against making any financial transactions on public Wi-Fi. Some even advise against checking social networks or email accounts for the same reason, because too much information can be exposed to hackers that can allow them to gain control of bank accounts. Also, there are rogue hot spots that direct users to legitimate-looking websites that prompt them to provide banking credentials.

Do not click on links that offer billion-dollar prizes

Everyone is familiar with emails saying, "Your email address has been selected to claim the sum of US\$ 500,000 in the 2015 European lottery." Expressions such as "your email address was selected" or "your address has won" are blabbermouth signs that the message is part of a scam. After all, you have not used your address to participate in a prize draw, have you? And if you have, it was unlikely to have been European lottery. Fraudsters obviously expect some recipients to suspect a scam and attempt to convince them otherwise.

Similarly, you may receive phishing messages promising a lottery win from Coca Cola, Google's anniversary winning notification, Yahoo lottery award international programme, Microsoft's award promotion and what not. Should you receive an email of this type, visit the specific company's official website; most likely you will find that the company is not actually holding a lottery of any kind. Google translate



service has made life much easier for online fraudsters as now they can send messages to users all over the world in various languages.

Do not trust customer service seeking banking credentials

A phishing mail is an email fraud method in which the perpetrator sends out legitimate-looking emails in an attempt to gather personal and financial information from recipients. Voice phishing (vishing) is the criminal practice of using social engineering over the telephone system to gain access to private personal and financial information from users with a financial motive. Some fraudsters use facilities like Voice over Internet Protocol, caller ID spoofing and automated systems (IVR).

Just like phishing, smishing uses mobile phone text messages to lure consumers. Often the text contains a URL or phone number. The phone number often has an automated voice response system. And again, just like phishing, the smishing message usually asks for your immediate attention. Do not respond to any such messages.

The questions you need to ask yourself are:

1. Do you know the sender of the email? If yes, still be cautious before clicking on a link. If no, do not click on any link.
2. Is there any attachment in the email? If yes, is it executable (a file with extensions like .exe, .bat,